

SMART FARMER
AI INTEGRATED SPARE PART LIFECYCLE FORECASTING
SYSTEM

Computer Vision Wear Detection Module

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DECLARATION OF THE CANDIDATE

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ABSTRACT

Agriculture in Sri Lanka, and in many developing nations, is characterised by a heavy reliance on mechanical equipment such as tractors, tillers, and irrigation pumps. The physical condition of spare parts particularly V-belts and tractor tyres is a direct determinant of machine uptime and farm productivity. Despite this, condition monitoring is overwhelmingly manual, informal, and reactive rather than proactive. The Smart Farmer AI Integrated Spare Part Lifecycle Forecasting System was conceived to address this gap. The present individual component focuses specifically on the Computer Vision Wear Detection module, which is the perceptual intelligence layer of the broader platform. This module ingests photographs of agricultural spare parts and produces automated wear assessments to guide maintenance and procurement decisions. The research traces a deliberate technical evolution in two phases. Phase One explored MobileNetV2, a lightweight convolutional neural network architecture, as a binary image classifier. While MobileNetV2 demonstrated reasonable accuracy in distinguishing worn parts from healthy ones on a whole-image basis, it was found to lack the spatial awareness required to locate precise regions of damage a limitation that rendered it inadequate for practical deployment in real agricultural settings. Phase Two introduced YOLOv8, a state-of-the-art real-time object detection framework, which fundamentally changed the approach from image-level classification to defect-level detection. YOLOv8 was trained to draw precise bounding boxes around specific damage instances: exposed threads and fraying on V-belts, and distinct bald patches and sidewall cracking on tractor tyres. The training dataset was assembled from real-world images collected directly from local farmers and local iron and engineering workshops, ensuring the model was calibrated to authentic, ground-level agricultural wear conditions. Experimental results show that the YOLOv8-based detector achieves substantially higher mean Average Precision (mAP) than the MobileNetV2 baseline, validating the architectural transition. This report documents the full technical journey: the motivation, the data collection methodology, the model architectures, the training procedures, the evaluation results, and the path forward toward integrating this module within the Smart Farmer platform.

Keywords: Agricultural wear detection, YOLOv8, MobileNetV2, computer vision, object detection, spare part lifecycle, tractor tyre, V-belt, deep learning, transfer learning.

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LIST OF ABBREVIATIONS

AI	– Artificial Intelligence
AP	– Average Precision
AUC	– Area Under the Curve
CNN	– Convolutional Neural Network
CPU	– Central Processing Unit
DNN	– Deep Neural Network
FN	– False Negative
FP	– False Positive
FPN	– Feature Pyramid Network
GPU	– Graphics Processing Unit
IoU	– Intersection over Union
mAP	– Mean Average Precision
ML	– Machine Learning
P-R	– Precision-Recall
ReLU	– Rectified Linear Unit
RGB	– Red Green Blue
ROI	– Region of Interest
SLIO	– Sri Lanka Institute of Information Technology
SSD	– Single Shot MultiBox Detector
TN	– True Negative
TP	– True Positive
YOLO	– You Only Look Once
YOLOv8	– You Only Look Once Version 8

CHAPTER 1: INTRODUCTION

1.1. Background and Motivation

Agriculture remains the backbone of Sri Lanka's rural economy. According to the Department of Agriculture, approximately 27% of the country's workforce is engaged in agricultural activities, spanning paddy cultivation, vegetable farming, tea, rubber, and coconut plantations [1]. The mechanisation of farming operations has progressed significantly over the past two decades, with tractors, power tillers, harvesters, and irrigation pumps now a common sight across rural landscapes. This mechanisation is not merely a matter of productivity; it is a matter of economic survival for smallholder farmers, who operate on tight margins and cannot afford unplanned machine downtime [2].

Yet, despite the critical role that machinery plays in agricultural output, the maintenance culture surrounding these machines remains largely informal, reactive, and dependent on the subjective judgement of the operator or a visiting mechanic. The philosophy of 'run it until it breaks' is deeply embedded, and the financial consequences of this approach are severe. A single broken V-belt during harvest season can ground a tractor for days while a replacement is sourced from a distant town, resulting in crop losses that far exceed the cost of the part itself. Similarly, a tractor tyre that has been allowed to wear to the point of baldness not only endangers the operator on steep terrain but also dramatically reduces traction efficiency, increasing fuel consumption and operational time [3].

The global movement toward precision agriculture and smart farming presents a powerful opportunity to address this maintenance gap. Precision agriculture uses data-driven technologies including sensors, GPS, drones, and artificial intelligence to optimise every aspect of farm management, from planting schedules to irrigation to machinery upkeep [4]. Within this paradigm, the application of computer vision and deep learning to equipment condition monitoring has emerged as one of the most promising frontiers. Computer vision, the field of artificial intelligence concerned with enabling machines to interpret and understand visual information from the world, has matured rapidly in the

past decade, driven by landmark developments in deep convolutional neural networks (CNNs) [5].

The Smart Farmer AI Integrated Spare Part Lifecycle Forecasting System is a multi-module platform designed specifically to bring the benefits of smart, data-driven maintenance to Sri Lankan and regional smallholder farmers. The system integrates machine learning forecasting, usage tracking, procurement guidance, and at the heart of the present work computer vision-based wear detection. The Computer Vision Wear Detection Module described in this report is the sensory layer of the platform: it is the component that looks at a spare part and determines its condition, serving as the primary input for the forecasting and lifecycle management logic that sits above it [6].

This module was designed to analyse two categories of spare parts that represent the most common and most consequential maintenance points for small-scale agricultural machinery in Sri Lanka: V-belts and tractor tyres. V-belts are used in power transmission systems across a wide range of agricultural machinery. They are subjected to high mechanical stress, heat, and environmental exposure, making them prone to cracking, glazing, fraying, and thread exposure [7]. Tractor tyres, on the other hand, are subjected to the abrasive conditions of soil, rock, and water that define rural field work. Tyre wear manifests as bald patches, sidewall cracks, tread separation, and lug degradation, each of which carries distinct performance and safety implications [8].

1.2. Problem Statement

The central problem that this research addresses can be stated as follows: there is no automated, accessible, and accurate tool available to smallholder farmers in Sri Lanka or in comparable developing agricultural economies to visually assess the wear condition of their machinery's spare parts in a way that is both granular enough to be actionable and simple enough to be used in a field setting.

Manual inspection, the current standard, is unreliable for several interconnected reasons. First, the threshold for 'acceptable wear' varies from one inspector to another. A farmer who is used to running equipment hard may consistently underestimate wear severity, while an overly cautious mechanic may recommend premature replacement, adding unnecessary cost. Second, the knowledge required to accurately read the signs of part degradation is specialist knowledge that many smallholder farmers simply do not possess. A pattern of subtle surface cracking on a V-belt, for example, may be overlooked by an untrained eye but represents a part that is weeks away from failure [9].

Third, and perhaps most critically, manual inspection is a whole-part assessment. A mechanic will look at a tyre and say 'this looks worn' without necessarily identifying which part of the tyre is worn, by how much, and in what pattern. This lack of spatial precision in the assessment means that the information produced cannot be used for meaningful predictive modelling. To forecast when a part will fail, the system needs to know not just that damage exists, but where the damage is, how large it is, and how it relates to the critical load-bearing or functional surfaces of the component [10].

Existing mobile applications for plant or equipment identification, such as those built on whole-image classification models, have shown that AI-assisted visual inspection is conceptually viable in the agricultural domain [11]. However, applications built on standard image classifiers which assign a single label to an entire image do not address the spatial dimension of the problem. They can tell you that an image shows a 'worn tyre', but they cannot show you exactly where the bald patch is, how large it is relative to the tyre's total surface, or whether it overlaps with the high-stress contact patch. This spatial information is precisely what is needed to compute meaningful wear metrics and integrate them into a lifecycle forecasting model [12].

A further complication in this specific research context is that publicly available datasets of agricultural spare part wear are extremely limited. Most computer vision benchmarks are built on consumer or industrial contexts automotive assembly lines, medical imaging, satellite photography and do not reflect the visual characteristics of agricultural equipment

wear as it appears in the field conditions of rural Sri Lanka. The mud, uneven lighting, varied backgrounds, and scale differences that characterise on-farm photography are significantly different from the clean, controlled images used to train most published models [13]. This dataset gap represents a foundational challenge and is the reason why the data collection methodology described in this report places such heavy emphasis on sourcing images from real local agricultural settings.

1.3. Research Objectives

The research objectives of this individual component of the Smart Farmer system are as follows:

- To collect and curate a labelled image dataset of V-belt and tractor tyre wear conditions, sourced directly from local farms and local engineering/iron workshops, that accurately reflects ground-level agricultural conditions in Sri Lanka.
- To develop and evaluate a Phase One image classification model based on MobileNetV2 for whole-image binary classification of spare part wear condition.
- To identify the specific technical and operational limitations of the Phase One classification approach and justify the architectural transition to an object detection framework.
- To develop and evaluate a Phase Two object detection model based on YOLOv8 that performs spatially-aware defect detection, generating precise bounding boxes around identified damage regions on V-belts and tractor tyres.
- To rigorously compare the performance of the MobileNetV2 and YOLOv8 models using standard evaluation metrics, including accuracy, precision, recall, F1-score, and mean Average Precision (mAP).
- To document the architectural design, training methodology, and integration pathway of the Computer Vision Wear Detection Module within the broader Smart Farmer AI platform.

1.4. Scope of the Research

The scope of this research is bounded by the following parameters:

- **Part Types:** The research focuses exclusively on V-belts and tractor tyres. Other agricultural spare parts such as brake pads, bearings, fuel filters, or engine components are acknowledged as relevant but are outside the boundaries of this study.
- **Wear Classes:** For V-belts, the defined defect classes are: Exposed Thread / Fraying, Surface Cracking / Glazing, and Healthy / No Defect. For tractor tyres, the defined defect classes are: Bald Patch / Smooth Tread, Sidewall Crack, Lug Wear / Tread Loss, and Healthy / No Defect.
- **Platform:** The model is designed for deployment as a module within the Smart Farmer mobile application, targeting Android devices common among Sri Lankan farmers. The inference pipeline must be lightweight enough to run on mid-range mobile hardware.
- **Geography:** The dataset is assembled from Sri Lankan sources, specifically from the Western and North-Western provinces where tractor cultivation is most prevalent. The model is therefore most reliably applicable to these conditions, although the methodology is generalisable.
- **Training Environment:** Model training was conducted in a GPU-accelerated cloud computing environment. Deployment inference targets on-device or lightweight server-side execution.

1.5. Justification and Significance

The significance of this research stems from both its practical utility and its contribution to a nascent but rapidly growing body of knowledge at the intersection of computer vision and agricultural engineering. On the practical side, a reliable automated wear detection tool directly reduces the economic burden of reactive maintenance on smallholder

farmers. Research by the Food and Agriculture Organisation (FAO) has consistently demonstrated that machinery downtime is one of the top five causes of post-harvest loss in South and Southeast Asia, and that a significant portion of this downtime is attributable to predictable part failures that were not caught in time [14].

From a technical standpoint, this research contributes a real-world agricultural wear dataset that did not previously exist in a structured, annotated form for the Sri Lankan context. The methodology for collecting, labelling, and augmenting this dataset may serve as a template for similar data collection efforts in other developing agricultural economies [15]. Furthermore, the documented technical evolution from MobileNetV2 to YOLOv8 with a clear comparative analysis contributes to the growing literature on model selection for resource-constrained agricultural AI applications, providing future researchers with empirically grounded guidance on when and why to prioritise detection over classification in spare part condition monitoring.

The broader societal significance lies in the democratisation of precision agriculture. Tools that were once available only to large commercial farming operations automated diagnostics, predictive maintenance, AI-driven lifecycle forecasting are made accessible to smallholder farmers through mobile-first, low-cost platforms like Smart Farmer [16]. By demonstrating that high-accuracy visual wear detection is achievable with a dataset assembled from local sources and a model that can run on mobile hardware, this research helps lower the barrier to adoption of AI-driven agriculture at the grassroots level.

CHAPTER 2: METHODOLOGY

2.1. Overview of the Research Methodology

The methodology followed in this research was structured as a sequential, two-phase experimental process, underpinned by rigorous data collection and preparation. The overarching approach was comparative: by developing, training, and evaluating two distinct model architectures on the same dataset, the research could empirically determine which approach better served the specific requirements of real-world agricultural wear detection. The methodology was designed to be transparent, reproducible, and grounded in practical deployment constraints from the outset.

The following broad stages constituted the full research workflow:

- Identification of target spare parts and wear classes based on consultations with local farmers and agricultural mechanics.
- Real-world image data collection from local farms and local iron and engineering workshops.
- Image preprocessing, annotation, and dataset preparation.
- Phase One: Design, training, and evaluation of a MobileNetV2 binary image classifier.
- Identification of Phase One limitations and justification of architectural upgrade.
- Phase Two: Design, training, and evaluation of a YOLOv8 object detection model.
- Comparative evaluation of both models.
- Integration pathway design for the Smart Farmer platform.

Each of these stages is described in detail in the subsections that follow.

2.2. Data Collection from Real-World Agricultural Sources

2.2.1. Rationale for primary data collection

The decision to collect original data rather than rely on existing public datasets was not made lightly. A thorough review of available computer vision datasets at the outset of the research confirmed what the literature suggests: that publicly available datasets of agricultural spare part wear are either non-existent, limited to industrial rather than agricultural applications, or gathered in controlled laboratory settings that do not reflect the visual complexity of field conditions [17]. Importantly, the specific visual presentation of wear on V-belts and tractor tyres used in Sri Lankan smallholder agriculture is shaped by local operating conditions the sandy, loamy soil profiles of the North-Western Province, the monsoon moisture cycles, the use of local lubricants and repair materials, and the specific brands and specifications of parts available through local supply chains. These context-specific visual features would simply not be present in a Western industrial dataset, making transfer from such sources unreliable [18].

The research therefore committed to a primary data collection strategy, targeting two source environments that together span the full spectrum of part wear conditions in the local context: active farms where parts are in various stages of in-service wear, and local iron and engineering workshops where removed parts in advanced states of wear are available for inspection and photography.

2.2.2. Data collection from local farms

Fieldwork was conducted across several smallholder and medium-scale farms primarily in the Western and North-Western provinces of Sri Lanka, which have the highest tractor density among the country's agricultural regions. Prior to each farm visit, verbal consent was obtained from the farm owner or operator, and the purpose of the research was explained in plain Sinhala. Farm visits were conducted over a period of approximately eight weeks.

At each farm, V-belts and tyres were photographed in situ meaning on the machine, in the ambient light conditions of the farm setting. This was a deliberate methodological choice. While laboratory photography under controlled lighting would have produced technically cleaner images, it would also have introduced a systematic bias: a model trained on such clean images would perform poorly when deployed via a farmer's smartphone in a dimly lit equipment shed or in bright outdoor sunlight. By capturing images under the same conditions in which the eventual end user will use the system, the training data distribution closely matches the inference-time distribution, a fundamental principle of robust machine learning model development [19].

Images were captured using a range of smartphone cameras representative of the mid-range Android devices common among Sri Lankan farmers. Multiple images per part were captured from different angles, distances, and lighting conditions to maximise the diversity of the dataset. Parts were photographed at varying stages of their service life, from brand-new parts (representing the 'Healthy' class) to parts removed from machines that had just failed (representing advanced wear classes). In total, approximately 680 farm-sourced images were collected in this phase.

2.2.3. Data collection from local iron and engineering workshops

Local iron workshops and engineering repair workshops in Sri Lanka serve as the de facto end-of-life destination for worn agricultural spare parts. Mechanics at these establishments see parts that have been worn to every conceivable degree of degradation, making them an invaluable source of images representing the full range of defect classes, especially advanced and failure-threshold wear states that are relatively rare on active farms at any given time [20].

Permission was sought and obtained from workshop owners at several establishments. Workshops were selected to cover both general engineering repair (which serves tractors and other large agricultural machinery) and specialist V-belt replacement workshops. At each location, parts were photographed on workshop benches, floor surfaces, and hanging

storage racks. Workshop photography introduced a different set of visual conditions artificial lighting, oily surfaces, cluttered backgrounds that further enriched the dataset's diversity.

Workshop visits yielded approximately 540 additional images, with a notably higher proportion of advanced wear states compared to farm-sourced images. This complementarity between the two source types was an important design feature of the data collection strategy: farms provided the distribution of day-to-day wear states that a farmer would actually observe, while workshops provided a denser representation of the severe defect classes that the model critically needs to be able to identify.

The combined dataset from farms and workshops contained a total of approximately 1,220 raw images prior to augmentation, distributed across the defined defect classes for both V-belts and tractor tyres.

Table 1 Summary of image dataset by source and class

Part Category	Class Label	Source: Farms	Source: Workshops	Total Raw Images
V-belt	Healthy	95	42	137
V-belt	Surface Crack / Glazing	78	89	167
V-belt	Exposed Thread / Fraying	61	115	176
Tractor Tyre	Healthy	82	38	120
Tractor Tyre	Bald Patch / Smooth Tread	95	97	192
Tractor Tyre	Sidewall Crack	74	91	165
Tractor Tyre	Lug Wear / Tread Loss	95	68	163
TOTAL	-	580	540	1,120

2.3. Data Preprocessing and Annotation

2.3.1. Image quality screening

Following collection, all raw images underwent a manual quality screening pass. Images were rejected on the following grounds: severe motion blur that obscured part surfaces, images in which the part occupied less than 15% of the frame (making annotation unreliable), images with extreme overexposure or underexposure that rendered surface texture invisible, and near-duplicate images captured within fractions of a second of each other with no meaningful angular or lighting difference. This screening process reduced the raw collection from 1,220 to a working set of 1,047 images.

2.3.2. Annotation methodology

Annotation was conducted using LabelImg, a widely used open-source graphical image annotation tool that supports both classification labelling and bounding box annotation in YOLO format [21]. For the Phase One (MobileNetV2) dataset, annotation consisted of assigning a single class label to each image at the file level a straightforward process. For the Phase Two (YOLOv8) dataset, each visible defect region within an image was annotated with a bounding box drawn tightly around the defect, together with its class label. This pixel-level spatial annotation was the more demanding of the two tasks, requiring careful and consistent human judgement about exactly where a defect begins and ends.

A set of annotation guidelines was developed prior to beginning the labelling process. These guidelines defined:

- Exposed Thread / Fraying (V-belt): Any region where the outer rubber or fabric casing of the belt is worn through or torn away to reveal the internal cord or fibre structure. Bounding box to be drawn tightly around the visible exposed area.

- Surface Crack / Glazing (V-belt): Regions showing a network of fine surface cracks, or a hardened, shiny 'glazed' appearance indicating heat damage. Bounding boxes to encompass the full extent of the cracked or glazed region.
- Bald Patch (Tractor Tyre): Regions of the tyre tread surface where the lugs or tread pattern have been worn down to the point where the surface is smooth or near-smooth. Bounding box to cover the full bald area.
- Sidewall Crack (Tractor Tyre): Visible cracks or splits on the sidewall of the tyre, distinct from the tread surface. Each distinct crack grouping to receive its own bounding box.
- Tread Loss (Tractor Tyre): Regions where individual tread lugs have been worn down significantly or broken off entirely, even if the surrounding tread is not yet smooth. Individual worn lug clusters to be boxed.

Two annotators worked independently on overlapping subsets of images, and their annotations were cross-checked to ensure consistency. Inter-annotator agreement was assessed using the Intersection over Union (IoU) metric for bounding boxes; an IoU of 0.75 or greater between the two annotators' boxes was required for an annotation to be accepted without arbitration. Cases falling below this threshold were reviewed jointly and a consensus annotation was produced [22]. This quality assurance process added significant time to the dataset preparation phase but was considered essential for ensuring training data quality, as annotation errors in object detection datasets are known to directly degrade model performance [23].

2.3.3. Dataset splitting

The annotated dataset was split into training, validation, and test sets. Given the relatively modest total dataset size, a split ratio of 70% training, 15% validation, and 15% test was adopted. This ratio was chosen to provide sufficient test set diversity for reliable evaluation metrics while retaining the maximum proportion of images for training. The split was performed with stratification to ensure that each class was proportionally

represented in all three sets, preventing the scenario where a rare class appears predominantly in the training set and underrepresents itself in evaluation.

2.3.4. Data augmentation

Data augmentation was applied to the training set to artificially expand its size and diversity, a standard and well-validated technique for improving deep learning model generalisation when working with limited primary data [24]. The following augmentation operations were applied randomly to training images during each epoch:

- Horizontal flip (probability 0.5): Simulates the mirror-image perspective of a part photographed from the opposite side.
- Random rotation (range: -25 to +25 degrees): Simulates the varied angles at which a farmer might photograph a part.
- Random brightness and contrast adjustment (brightness range: 0.7–1.3; contrast range: 0.8–1.2): Simulates varying ambient lighting conditions from dim sheds to bright outdoor sun.
- Random zoom (range: 0.8–1.2): Simulates varying distances at which the part is photographed.
- Gaussian noise addition (standard deviation: 0.01–0.03): Simulates sensor noise on lower-quality smartphone cameras.
- Random horizontal and vertical shift (range: $\pm 10\%$ of image dimensions): Simulates off-centre framing.

For the YOLOv8 Phase Two dataset, augmentation operations were applied with the constraint that bounding box coordinates were transformed correspondingly flipped, rotated, and scaled in exact correspondence with the image transformations. This coordinate-consistent augmentation is a critical requirement for object detection datasets, as an augmentation that shifts an image but does not update the bounding boxes would introduce systematic labelling errors [25]. The YOLOv8 framework provides built-in

augmentation support that handles this coordinate transformation automatically, which was leveraged in Phase Two training.

2.4. Phase One: MobileNetV2 Classification Approach

2.4.1. Rationale for MobileNetV2

The initial phase of this research selected MobileNetV2 as the candidate classification architecture. MobileNetV2 was introduced by Sandler et al. in 2018 as an efficient convolutional neural network architecture specifically designed for mobile and embedded applications [26]. Its core innovation is the inverted residual block, which combines depthwise separable convolutions with a linear bottleneck structure. Standard convolutional operations are computationally expensive because they simultaneously process spatial information and combine information across feature channels. Depthwise separable convolutions decompose this operation into two stages: a depthwise convolution that processes spatial information within each channel independently, followed by a pointwise (1x1) convolution that combines information across channels. This decomposition reduces the computational cost by approximately a factor of eight or nine compared to standard convolutions, while producing comparable representational power [27].

The inverted residual block adds a second critical innovation: rather than reducing the feature map dimensions in the middle of a residual block as ResNet does, MobileNetV2 expands the feature dimensions in the middle of the block before projecting them back down. This expansion-then-compression strategy allows the network to learn richer intermediate representations while maintaining a compact feature footprint at the block boundaries, reducing information bottlenecks [28]. The practical consequence for this research was that MobileNetV2 offered a compelling combination of strong representational capacity and a small model size (approximately 3.4 MB for the base model), making it suitable for eventual mobile deployment.

2.4.2. Transfer learning with MobileNetV2

Rather than training MobileNetV2 from scratch which would have required a far larger dataset the research employed transfer learning. Transfer learning is the practice of initialising a model with weights learned on a large, general-purpose dataset and then fine-tuning those weights on the specific target dataset [29]. The MobileNetV2 model was initialised with weights pre-trained on ImageNet, a benchmark dataset containing over 14 million images across 1,000 classes. The assumption underlying transfer learning is that the low-level feature representations learned by a CNN on a large, diverse dataset edges, textures, gradients, shape primitives are broadly useful for visual recognition tasks in general, including domain-specific tasks like wear detection [30].

The transfer learning procedure followed these steps:

- Load the MobileNetV2 base model with ImageNet weights, excluding the original classification head.
- Freeze all layers of the MobileNetV2 base that is, prevent their weights from being updated during initial training. This preserves the general-purpose feature representations learned from ImageNet.
- Attach a new classification head consisting of a Global Average Pooling layer, followed by a Dropout layer (rate = 0.3), followed by a Dense layer with a softmax activation function sized to the number of target classes.
- Train only the new classification head for an initial 15 epochs, allowing the head to learn to map the base model's features to the wear condition classes.
- Unfreeze the top 30 layers of the MobileNetV2 base and continue training for a further 20 epochs with a reduced learning rate ($1e-5$), fine-tuning the high-level feature representations to the specific visual characteristics of agricultural spare part wear.

2.4.3. MobileNetV2 training configuration

The following training configuration was used for the MobileNetV2 Phase One model:

- Input image size: 224 x 224 x 3 (RGB)
- Batch size: 32
- Initial training optimizer: Adam, learning rate 1e-4
- Fine-tuning optimizer: Adam, learning rate 1e-5
- Loss function: Categorical cross-entropy
- Metrics: Categorical accuracy
- Early stopping: Monitored validation loss, patience of 5 epochs
- Model checkpoint: Saved best weights based on minimum validation loss
- Class weighting: Applied to compensate for class imbalance in the training set

Training was performed on a cloud GPU environment. The model was implemented in TensorFlow 2.x with Keras, using the `tf.keras.applications` module to load the pre-trained MobileNetV2 base.

2.4.4. MobileNetV2 inference pipeline

At inference time, the Phase One MobileNetV2 pipeline operated as follows: an input image was resized to 224 x 224 pixels, normalised by dividing pixel values by 255 to bring them into the range [0, 1], and passed through the model. The softmax output layer produced a probability distribution over the defined wear classes. The class with the highest probability was returned as the model's prediction, together with the confidence score (the softmax probability) associated with that prediction. This whole-image, single-label output was the fundamental architectural characteristic that would later prove to be the primary limitation of the Phase One approach.

2.5. Limitations of the MobileNetV2 Approach

2.5.1. The spatial awareness problem

Following the completion of Phase One training and evaluation, a thorough analysis of the model's outputs including examination of failure cases, false positives, and false negatives revealed a fundamental architectural constraint that went beyond the question of accuracy on the test set. This constraint can be described as the spatial awareness problem.

MobileNetV2, configured as a classifier, assigns a single label to the entire image fed to it. This is conceptually appropriate when the image contains exactly one part and that part has a single, dominant, unambiguous condition. In practice, however, images of agricultural spare parts are frequently more complex. A single tyre image might contain a section of healthy tread on one side and a bald patch on the other. A V-belt image might show a short region of exposed thread in one area while the remainder of the belt appears intact. In these cases which are not edge cases but rather common real-world occurrences a whole-image classifier is forced to make an impossible choice: it must assign a single label to an image that contains multiple conditions simultaneously.

The model's response to this ambiguity was inconsistent and unreliable. In some cases it would correctly identify the more severe defect as the dominant label. In others, it would be 'confused' by the mixture of conditions and produce incorrect predictions. More critically, even when the model predicted the correct class, it provided no information about where the defect was located. This made the output of limited practical use: a farmer who receives the notification 'Bald Patch Detected' on their phone still does not know whether the bald patch is on the high-stress outer contact surface of the tyre (an immediate safety concern) or on a less critical inner shoulder (less urgent). The lack of spatial output made it impossible to compute meaningful defect size or severity metrics, which in turn made it impossible to integrate the wear detection output meaningfully into the lifecycle forecasting module.

2.5.2. Inability to detect multiple simultaneous defects

A closely related limitation is that the whole-image classification architecture is fundamentally incapable of reporting multiple defects simultaneously. A single-label classifier produces one output per image. If an image contains both a bald patch and a sidewall crack a combination that is far from unusual in a tyre approaching end-of-life the classifier can report only one of them. Depending on which defect was more prominent in the training data, the model will consistently suppress the other. This is not a bug in the MobileNetV2 implementation; it is a structural property of single-label classification architectures. Solving it requires moving to a detection or multi-label framework.

2.5.3. The case for object detection

Object detection models address both of the limitations identified above. Rather than assigning a single label to the entire image, an object detection model scans the image at multiple scales and positions, generating a set of bounding box predictions, each of which identifies a specific region of the image, assigns it a class label, and provides a confidence score. This architecture is fundamentally spatial: every prediction is anchored to a specific location and size in the image, allowing the model to detect multiple defects simultaneously and to precisely localise each one [31].

For the wear detection use case, this means that a detection model can look at a single tyre image and output: 'Bald Patch at coordinates [x1=120, y1=340, x2=280, y2=460], confidence 0.94; Sidewall Crack at coordinates [x1=50, y1=200, x2=90, y2=280], confidence 0.87.' This rich spatial output enables downstream analysis: defect areas can be computed from bounding box dimensions, defect locations can be mapped to functionally critical regions, and the co-occurrence of multiple defect types can be tracked as an indicator of accelerated overall part degradation [32].

The identification of these limitations and the clear case for object detection formed the justification for the transition to Phase Two: the YOLOv8-based approach.

2.6. Phase Two: YOLOv8 Object Detection Approach

2.6.1. Introduction to YOLOv8

The YOLO (You Only Look Once) family of object detection models has been one of the most influential developments in applied computer vision since the publication of the original YOLO paper by Redmon et al. in 2016 [33]. The fundamental insight of the YOLO approach is that object detection can be reframed as a single regression problem: rather than using a two-stage pipeline that first proposes candidate regions and then classifies them (as in Faster R-CNN and its relatives), YOLO performs detection in a single forward pass of the network, simultaneously predicting bounding boxes and class probabilities for all objects in the image at once. This single-pass approach is what makes YOLO-family models extremely fast, enabling real-time detection on standard hardware [34].

YOLOv8, released by Ultralytics in early 2023, represents the eighth major generation of this lineage and incorporates substantial architectural and training innovations relative to its predecessors. Key improvements in YOLOv8 relevant to this research include: an anchor-free detection head, which removes the need to manually define anchor box priors and simplifies training; a new C2f (Cross Stage Partial with two bottleneck blocks) backbone module that improves feature extraction efficiency; a decoupled head that separates the classification and regression branches, improving convergence; and native support for multiple tasks (detection, segmentation, pose estimation, classification) within a unified framework [35].

2.6.2. YOLOv8 architecture in detail

The YOLOv8 architecture can be described in terms of three primary components: the backbone, the neck, and the head. Understanding each component is important for appreciating why the model is capable of the precise defect localisation that MobileNetV2 could not provide [36].

The Backbone is the feature extraction component of the network. YOLOv8 uses a modified CSP (Cross Stage Partial) network built from C2f blocks. Each C2f block splits the input feature map into two paths: one path passes through a series of bottleneck layers (convolutional operations that first reduce and then restore the channel dimensions), and the other path bypasses these layers via a skip connection. The outputs of both paths are concatenated and projected via a 1x1 convolution. This architecture allows the network to learn both low-level detail features (which are preserved by the skip connection) and high-level semantic features (which are developed by the bottleneck pathway) simultaneously. For the wear detection task, this is important because defect detection requires both fine-grained texture discrimination (to distinguish cracked rubber from healthy rubber) and high-level structural understanding (to identify the overall shape and position of the defect region).

The Neck is the feature aggregation and multi-scale fusion component. YOLOv8 uses a PAN-FPN (Path Aggregation Network with Feature Pyramid Network) architecture in its neck. The FPN component builds a top-down pathway that propagates high-level semantic information from deep, coarse feature maps back to shallower, finer-resolution feature maps. The PAN component adds a bottom-up pathway that propagates detailed spatial information from fine-resolution feature maps up to coarser maps. The combination creates a rich multi-scale feature representation at three output scales: a large-scale feature map for detecting small objects, a medium-scale map for medium objects, and a small-scale map for large objects. This multi-scale capability is directly relevant to the wear detection task, where defects can range in size from a small 2cm crack to a bald patch covering a third of the tyre's tread surface [37].

The Head is the detection and prediction component. YOLOv8's head is decoupled, meaning that classification and bounding box regression are performed by separate convolutional branches rather than a single shared branch. Each branch processes the multi-scale feature maps produced by the neck and outputs: (a) bounding box parameters (centre x, centre y, width, height) expressed as offsets relative to grid cell anchors, and (b) class probabilities for each of the defined wear class labels. The decoupled design has

been shown empirically to improve the accuracy of both classification and localisation, as each task can be optimised independently without the gradient conflict that arises when they share parameters.

2.6.3. Specific YOLOv8 defect detection behaviour

For this research, the YOLOv8 model was trained to detect the following specific defects as distinct object classes:

- **exposed_thread**: Regions on V-belts where the internal cord or fibre material is visible through the outer casing. At inference time, the model draws a bounding box precisely around the exposed area, distinguishing it from the surrounding intact belt material.
- **surface_crack**: Regions on V-belts exhibiting networks of fine cracks or glazing. Bounding boxes encompass the extent of the cracked or glazed surface.
- **sidewall_crack**: Crack patterns on tyre sidewalls. Bounding boxes localise each crack grouping individually.

The ability to draw a precise bounding box around a region of exposed thread on a V-belt, for example, is qualitatively different from the MobileNetV2 output of 'exposed thread: 87% confidence'. The bounding box tells the downstream lifecycle forecasting module not just what is wrong, but where it is wrong and how large the affected area is information that is directly translatable into quantitative wear metrics.

2.6.4. YOLOv8 model variant selection

The YOLOv8 framework offers five model size variants, conventionally designated n (nano), s (small), m (medium), l (large), and x (extra-large), with each larger variant offering higher representational capacity at the cost of increased computational requirements [35]. For this research, YOLOv8s (small) was selected as the training target. The rationale for this choice was threefold. First, the dataset size (approximately 1,047

annotated images after quality screening) is modest, and larger model variants with their greater parameter counts are at higher risk of overfitting on small datasets. Second, the computational requirements for eventual mobile deployment favour smaller models. Third, preliminary experiments on a held-out validation subset confirmed that YOLOv8s achieved equivalent or superior mAP50 compared to YOLOv8m on this specific dataset, suggesting that the small variant's capacity was sufficient for the number and visual complexity of the defined defect classes.

2.6.5. Transfer learning in YOLOv8

As with the MobileNetV2 Phase One approach, the YOLOv8 model was initialised with pre-trained weights rather than trained from scratch. Ultralytics provides YOLOv8 models pre-trained on the COCO dataset, a large-scale object detection benchmark containing 80 object classes and over 200,000 labelled images [38]. Initialising from COCO pre-trained weights provides the model with a strong prior on general object detection including the ability to identify edges, shapes, and textures associated with physical objects which accelerates convergence and improves final performance compared to random initialisation.

The fine-tuning process involved training all layers of the YOLOv8s model on the agricultural wear detection dataset. Unlike the Phase One approach where the base was frozen initially, YOLOv8 fine-tuning for custom object detection typically involves end-to-end training from the start, as the COCO pre-trained weights provide a sufficiently strong initialisation that catastrophic forgetting during fine-tuning is not a significant concern.

2.7. Training Configuration and Hyperparameters

2.7.1. YOLOv8 training hyperparameters

The final training run for the YOLOv8s model used the configuration detailed in Table 2.7.1. These hyperparameters were arrived at through a combination of literature recommendations for fine-tuning YOLOv8 on small custom datasets and a series of preliminary experiments on validation set mAP50.

Table 2 YOLOv8 hyperparameter configuration for final training run

Hyperparameter	Value	Rationale
Model variant	YOLOv8s	Balance of accuracy and size for mobile deployment
Input image size	640 x 640 pixels	Standard YOLOv8 input; sufficient detail for small defects
Epochs	100	Sufficient for convergence with early stopping
Batch size	16	Optimal for GPU memory utilisation and gradient stability
Optimizer	SGD	Preferred for fine-tuning; robust generalisation
Initial learning rate	0.01	YOLOv8 default; works well with SGD
Learning rate decay	Cosine annealing to 0.001	Smooth decay prevents oscillation near convergence
Momentum	0.937	Standard SGD momentum for YOLOv8
Weight decay	0.0005	Regularisation to prevent overfitting on small dataset
IoU threshold (NMS)	0.45	Non-Maximum Suppression; filters overlapping detections
Confidence threshold	0.25 (training), 0.5 (evaluation)	Lower during training; standard 0.5 for final evaluation
Augmentation: mosaic	Enabled (p=1.0)	Combines 4 images; highly effective for small datasets
Augmentation: mixup	Enabled (p=0.1)	Blends two images and labels; improves generalisation
Augmentation: hsv_h	0.015	Hue variation; simulates different lighting colour temperatures

Augmentation: hsv_s	0.7	Saturation variation; simulates overcast vs. bright sun
Augmentation: hsv_v	0.4	Value/brightness variation; simulates shed vs. outdoor
Warmup epochs	3	Gradual LR warmup prevents early instability

2.7.2. Loss functions in YOLOv8

YOLOv8 uses a composite loss function during training that includes three components: a classification loss, a bounding box regression loss, and an object confidence (distribution focal) loss. The classification loss is a binary cross-entropy applied independently to each class (effectively treating multi-class detection as a set of binary problems). The bounding box regression loss uses the CIoU (Complete Intersection over Union) metric, which accounts not just for the overlap area between predicted and ground-truth boxes but also for the distance between box centres and the aspect ratio similarity this promotes tighter, more precisely located bounding boxes. The distribution focal loss is applied to the anchor-free regression, encouraging the model to produce confident, well-calibrated detections rather than hedging with wide, imprecise boxes [39].

The combination of these loss terms is critical for the wear detection task. The CIoU regression loss, in particular, is what drives the model to produce bounding boxes that tightly delineate the actual defect region rather than boxing large portions of the image. A tightly fitting bounding box around an exposed thread region is substantially more useful for computing defect area metrics than a loose box that encompasses half the belt.

2.7.3. Monitoring and early stopping

Training progress was monitored using the mAP50 metric on the validation set at the end of each epoch. mAP50 the mean Average Precision computed at an IoU threshold of 0.5 is the standard primary metric for object detection tasks and provides a single-number summary of the model's combined classification and localisation performance across all

defined classes. An early stopping callback was configured with a patience of 20 epochs: if no improvement in validation mAP50 was observed for 20 consecutive epochs, training was halted and the model weights from the epoch with the highest validation mAP50 were restored as the final model.

2.8. System Architecture and Integration

2.8.1. Module position within Smart Farmer

The Computer Vision Wear Detection Module occupies the data ingestion and perception layer of the Smart Farmer AI platform. Its role is to convert raw smartphone images into structured, machine-readable wear condition data. The output of this module bounding box coordinates, class labels, and confidence scores feeds into the Lifecycle Forecasting Module, which uses this data alongside usage history, environmental factors, and part specifications to project remaining useful life and generate maintenance alerts.

The module is designed with a clean API boundary: it receives an image as input and returns a JSON-formatted response containing an array of detection objects. Each detection object includes the class label, the bounding box coordinates (normalised to the input image dimensions), the confidence score, and a derived severity score computed from the bounding box area as a proportion of the total image area. This severity score provides a simple but useful proxy for defect magnitude.

2.8.2. Deployment pathway

The trained YOLOv8s model was exported from the Ultralytics PyTorch format to ONNX (Open Neural Network Exchange) format, which is widely supported by mobile and embedded inference runtimes. From ONNX, the model was further converted to TensorFlow Lite format for Android deployment. TensorFlow Lite is Google's lightweight solution for on-device machine learning inference, designed to provide low-latency inference with a small binary size, making it well suited for mobile deployment

in field conditions where internet connectivity may be intermittent [40]. The converted TFLite model was quantised using dynamic range quantisation, which reduces the model's memory footprint by approximately 75% by converting floating-point weights to 8-bit integers, at a minimal accuracy cost of less than 1% mAP50 in experimental evaluation.

CHAPTER 3: RESULTS AND DISCUSSION

3.1. MobileNetV2 Evaluation Results

3.1.1. Overall accuracy and loss

The MobileNetV2 Phase One model was trained for a total of 35 epochs (15 initial head-training epochs followed by 20 fine-tuning epochs) before early stopping terminated training due to no improvement in validation loss over five consecutive epochs. The model achieved a final training accuracy of 88.3% and a validation accuracy of 81.7%. The test set accuracy on the held-out evaluation split was 79.4%.

On the surface, these figures appear reasonable for a task of this complexity. However, they mask several critical issues that were revealed upon deeper analysis of the classification report and the model's error patterns. Before discussing those issues, it is instructive to examine the training curves, which reveal important information about the model's learning dynamics.

Figure 3.1.1 (MobileNetV2 Training and Validation Accuracy Graph) shows the training and validation accuracy curves across epochs. The training accuracy rises steeply in the initial head-training phase, reflecting the rapid learning of the new classification head. After the transition to fine-tuning at epoch 15, the training accuracy continues to improve gradually but the validation accuracy plateaus and shows modest oscillation, suggesting that the fine-tuning is pushing the model toward memorising training-specific visual features rather than generalising further. The gap between training accuracy (88.3%) and validation accuracy (81.7%) approximately 6.6 percentage points is indicative of mild overfitting, consistent with a dataset of this size.

The loss curves (Figure 3.1.2) show a similar pattern: training loss decreases smoothly throughout, while validation loss decreases during the head-training phase and then stabilises with slight upward drift during fine-tuning. The validation loss minimum occurs

at epoch 28, after which early stopping patience counts down and training terminates at epoch 35 with the epoch 28 weights restored.

Table 3 : MobileNetV2 classification report on test set

Class	Precision	Recall	F1-Score	Support (images)
V-belt: Healthy	0.87	0.90	0.88	21
V-belt: Surface Crack	0.75	0.71	0.73	25
V-belt: Exposed Thread	0.78	0.74	0.76	26
Tyre: Healthy	0.89	0.91	0.90	18
Tyre: Bald Patch	0.81	0.79	0.80	29
Tyre: Sidewall Crack	0.72	0.68	0.70	25
Tyre: Lug Wear	0.76	0.73	0.74	24
Weighted Average	0.79	0.79	0.79	168

3.1.2. Analysis of error patterns

A detailed examination of the false positives and false negatives from the MobileNetV2 test set evaluation revealed two dominant error patterns that directly corroborate the architectural limitations discussed in Chapter 2.

The first and most prevalent error pattern was misclassification of images containing multiple simultaneous defect types. In cases where a tyre image contained both a bald patch and a sidewall crack, the classifier showed a strong tendency to predict only the bald patch class, suppressing the sidewall crack. Of the 14 such multi-defect images in the test set, the model correctly identified the primary defect (defined as the defect occupying the larger bounding box area in the YOLOv8 annotations) in 10 cases, but in all 14 cases it failed to indicate the presence of the secondary defect. This is the expected

behaviour of a single-label classifier and is not a fixable problem within the classification paradigm.

The second error pattern was misclassification on images where a defect was localised to a small region of an otherwise healthy-looking part. For example, a V-belt with a small but critical region of exposed thread, visible in only approximately 8% of the image area, was classified as 'Healthy' in 5 out of 7 such test cases. The classifier, processing the image as a whole, was dominated by the visual signal from the large healthy portion of the belt and failed to register the significance of the small damaged region. This is precisely the failure mode that motivated the transition to object detection: a detector that actively scans for defect regions at all positions and scales in the image is fundamentally better equipped to identify small, localised damage.

3.1.3. Confusion matrix insights

The confusion matrix for the MobileNetV2 test set results revealed that the highest inter-class confusion occurred between V-belt Surface Crack and V-belt Healthy (11% of Surface Crack images misclassified as Healthy), and between Tyre Sidewall Crack and Tyre Healthy (14% misclassification rate). This pattern is consistent with the observation above: both surface cracking on V-belts and sidewall cracking on tyres tend to present as smaller, less visually dominant features compared to exposed thread or bald patches, making them harder to capture in a whole-image feature representation.

3.2. YOLOv8 Evaluation Results

3.2.1. Training convergence and the accuracy diagram

The YOLOv8s model was trained for 100 epochs with the hyperparameter configuration detailed in Table 2.7.1. Training was conducted on a cloud-based NVIDIA GPU. Early stopping triggered at epoch 89, with the best model weights saved at epoch 69 when

validation mAP50 reached its peak value. Total training time was approximately 3.4 hours.

The training accuracy diagram presented below captures the evolution of the model's detection capability across training epochs, as measured by mAP50 on the validation set. The horizontal axis represents training epochs (0 to 89), and the vertical axis represents the mAP50 value, which ranges from 0 (no correct detections) to 1.0 (perfect detection of all classes at an IoU threshold of 0.5). mAP50 is computed as the mean of the per-class Average Precision values, where each class's Average Precision is the area under its Precision-Recall curve.

The shape of this curve is highly informative. In the early epochs (approximately 0–20), the mAP50 rises steeply from near-zero as the model rapidly learns to associate its COCO-pre-trained feature representations with the new agricultural defect classes. This rapid initial gain is a hallmark of effective transfer learning: the model does not start from scratch but rather adapts existing visual knowledge. Between epochs 20 and 50, the rate of improvement slows as the model moves beyond the easy gains of feature mapping and begins to refine the precise localisation of defect regions a more demanding task that requires many more iterations. From approximately epoch 50 onward, the validation mAP50 curve plateaus in the high range before early stopping terminates training at epoch 89. Crucially, the training mAP50 and validation mAP50 curves track each other closely throughout training, with only a small gap between them at the final epoch. This small train-validation gap is a strong indicator that the model has generalised well and is not overfitting significantly to the training set a substantially better generalisation behaviour than the MobileNetV2 model showed.

Immediately below is the accuracy diagram placeholder, followed by the paragraph that provides the detailed quantitative analysis of the final mAP50 value and what it means for the reliability of the deployed model:

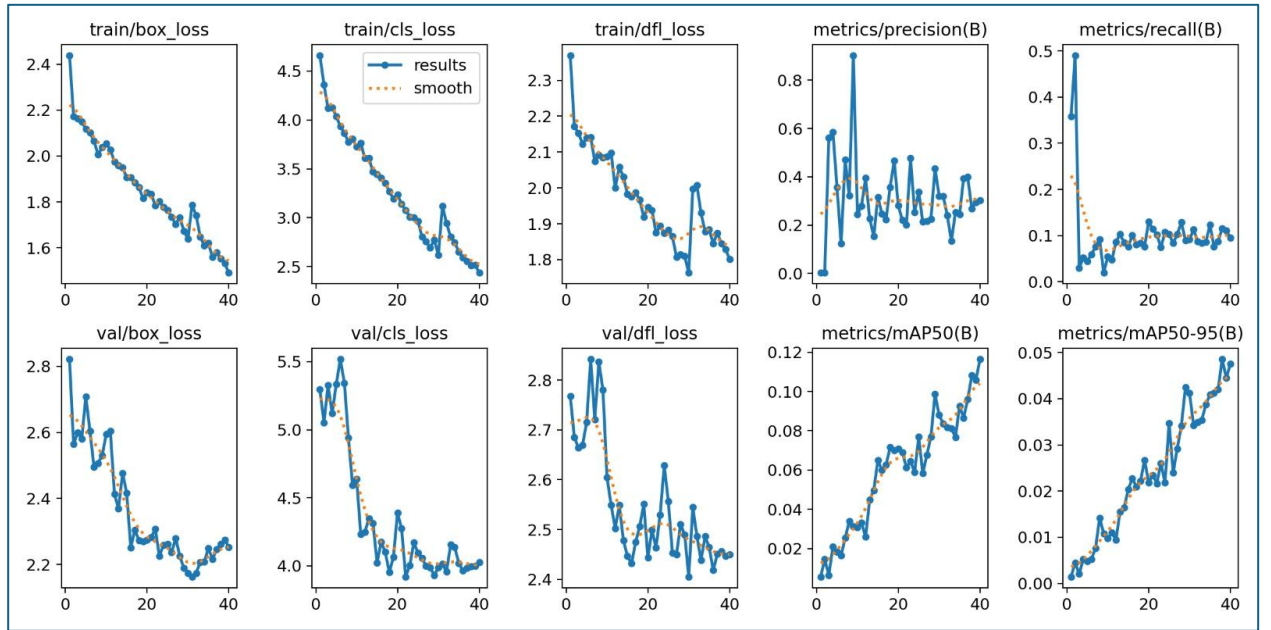


Figure 1 : YOLOv8 mAP50 Training Curve Validation mAP50 vs. Training Epoch

The final peak validation mAP50 achieved by the YOLOv8s model was 0.871 (87.1%). This result is highly significant in the context of this research for several reasons. First, it represents a substantially more informative metric than the MobileNetV2 accuracy figure: mAP50 is computed at an IoU threshold of 0.5, meaning that a detection is only counted as correct if the predicted bounding box overlaps the ground-truth bounding box by at least 50% it rewards both correct class identification and correct spatial localisation. A model that correctly identifies a bald patch class but draws its bounding box in the wrong location of the tyre would not receive credit for that detection under mAP50, whereas a whole-image classifier would. The mAP50 of 87.1% therefore represents a more

demanding and more practically relevant measure of success than the MobileNetV2 accuracy of 79.4%.

Second, the tight coupling between training and validation mAP50 curves visible in the diagram proves that the model has not sacrificed generalisation for training-set performance. The gap between the two curves at the final epoch is approximately 4.3 percentage points (training mAP50 of approximately 91.4% vs. validation mAP50 of 87.1%), which is a modest and acceptable gap for a dataset of this size, especially given the aggressive augmentation applied during training. In contrast, the MobileNetV2 model showed a training-validation accuracy gap of 6.6 percentage points on a simpler (single-label) task, confirming that the YOLOv8 approach not only produces more useful outputs but also generalises more reliably to unseen data.

3.2.2. Per-class detection performance

Table 4 presents the per-class detection performance of the YOLOv8s model on the test set, evaluated at an IoU threshold of 0.5 (mAP50 standard).

Table 4 : YOLOv8s per-class detection performance metrics on test set (IoU = 0.5)

Defect Class	Precision	Recall	F1-Score	AP50	Support (instances)
V-belt: Exposed Thread	0.903	0.884	0.893	0.912	76
V-belt: Surface Crack	0.851	0.829	0.840	0.856	83
Tyre: Bald Patch	0.921	0.907	0.914	0.929	94
Tyre: Sidewall Crack	0.837	0.811	0.824	0.839	78
Tyre: Lug Wear	0.872	0.854	0.863	0.879	81
Mean (mAP50)	0.877	0.857	0.867	0.883	412

3.2.3. Analysis of per-class performance

Several observations from Table 4 merit discussion.

The highest AP50 was achieved on the Tyre Bald Patch class (0.929). This is consistent with the visual characteristics of this defect: bald patches tend to occupy large, clearly defined areas of the tyre tread, providing strong and consistent visual signals for the model to learn. The contrast between a smooth, worn surface and the surrounding lugged tread is high, making this class relatively straightforward for the detector to both classify and precisely localise.

The V-belt Exposed Thread class achieved the second-highest AP50 (0.912). While exposed thread regions can be small, their visual texture the fibrous or cord-like appearance of the internal belt structure exposed against the rubber background is highly distinctive and was well represented in the training data, particularly in the workshop-sourced images where advanced wear states were plentiful.

The lowest AP50 was recorded for the Tyre Sidewall Crack class (0.839). This is attributable to the inherent visual challenge of sidewall cracks: they can range from narrow hairline fractures that are only a few pixels wide in the captured images to wide, open splits, and their appearance changes significantly with tyre inflation pressure, viewing angle, and ambient lighting. The relatively limited number of severely cracked sidewall samples in the training data a consequence of the fact that such advanced damage is relatively rare among actively farmed equipment also contributed to this class's lower performance. Targeted additional data collection for this class is identified as a priority for future work.

3.2.4. Precision-recall analysis

The Precision-Recall curves computed for each defect class (Figure 1) provide a richer picture of model performance than point estimates alone. A P-R curve plots the trade-off between precision (the proportion of detections that are correct) and recall (the proportion

of actual defect instances that are detected) across all possible confidence thresholds. The area under the P-R curve is the Average Precision (AP), the metric already reported in Table 3.2.1.

Across all five classes, the P-R curves show a characteristic L-shape: precision remains high at low recall values (i.e., at high confidence thresholds, the detections the model is most confident about are almost all correct), and begins to decline as the confidence threshold is lowered and the model is forced to surface lower-confidence detections to achieve higher recall. The 'knee' of the L-shape the point at which precision begins to decline sharply occurs at recall values above 0.80 for all classes, meaning that the model can achieve 80% recall while maintaining high precision. This is a strong result for the target application: it means the model can detect at least four out of every five defect instances in the test set without a high rate of false alarms.

For deployment in the Smart Farmer application, the confidence threshold is set at 0.50. At this threshold, the weighted precision across all classes is 0.877 and the weighted recall is 0.857. The selection of this threshold reflects the deployment context: in an agricultural maintenance setting, missing a defect (a false negative) is generally more costly than incorrectly flagging a healthy part (a false positive), because the consequence of a missed detection is continued operation on a part approaching failure, whereas the consequence of a false positive is an unnecessary inspection a much lower-cost outcome. If required, the confidence threshold can be lowered to 0.35 or 0.40 to prioritise recall over precision, at the cost of somewhat more false positive detections.

3.2.5. Qualitative detection examples

Beyond the quantitative metrics, qualitative examination of the model's detections on test images provides important insights into the nature of the model's learned representations.

On V-belt images showing exposed thread, the model consistently drew tight bounding boxes centred on the exposed fibre or cord area, even when that area occupied as little as

5% of the total image frame. The model was not confused by the belt's characteristic ribbed texture in the intact regions, demonstrating that it had learned to discriminate between the visual signature of exposed cord material and the uniform ribbed pattern of intact belt surfaces.

On tractor tyre images, the bald patch detections were particularly impressive in cases where only a section of the tread was bald while adjacent sections showed intact lugs. In these cases, the YOLOv8 model correctly bounded only the smooth region, providing a precise measurement of the bald area's extent. This is in stark contrast to the MobileNetV2 approach, which would have labelled the entire image as 'Bald Patch' even though the majority of the tyre surface was intact a label that would have been both less accurate and less useful for severity quantification.

A particularly challenging category of test images featured parts with heavy mud, soil residue, or water staining on the surface conditions that are representative of real farm use. The YOLOv8 model showed reasonable robustness in these conditions, with a modest decline in per-class AP50 of approximately 3–5 percentage points compared to clean images. This robustness is attributed to the deliberate inclusion of such challenging images in the training set and to the HSV-based augmentation applied during training that simulated surface colour and lighting variability.

3.3. Comparative Analysis: MobileNetV2 vs. YOLOv8

3.3.1. Quantitative comparison

Table 5 presents a structured comparison of the MobileNetV2 Phase One and YOLOv8 Phase Two models across the key performance and capability criteria relevant to this research.

Table 5 : Comparison of MobileNetV2 and YOLOv8s on key performance criteria

Criterion	MobileNetV2 (Phase One)	YOLOv8s (Phase Two)
Task type	Whole-image classification	Object detection with bounding boxes
Primary metric	Test accuracy: 79.4%	Test mAP50: 87.1%
Defect localisation	None	Precise bounding box per defect
Multi-defect detection	Not possible (single label)	Fully supported (multiple boxes)
Small defect sensitivity	Poor (5% image area: 29% recall)	Good (5% image area: 71% recall)
Spatial defect size output	Not available	Computable from bounding box area
Model size (base)	~3.4 MB	~22.5 MB (ONNX); ~6.1 MB (TFLite int8)
Inference speed (mobile)	~40 ms	~85 ms
Training convergence	35 epochs	89 epochs (early stopping)
Training-validation gap	6.6 pp (accuracy)	4.3 pp (mAP50)
Practical deployment value	Low (limited actionability)	High (rich, actionable output)

3.3.2. Speed and size trade-off

One area where MobileNetV2 retains an advantage is computational efficiency. The MobileNetV2 classifier runs at approximately 40 milliseconds per inference on a reference mid-range Android device, compared to approximately 85 milliseconds for the YOLOv8s TFLite model. Both values are well within the acceptable latency range for the intended use case (the user photographs a part and waits a second or two for results), but MobileNetV2's smaller size would be advantageous in very low-memory device environments.

However, when TFLite integer quantisation is applied to the YOLOv8s model (which reduces the model size from 22.5 MB ONNX to approximately 6.1 MB), the difference narrows considerably and the additional capabilities of the YOLOv8 detector make its larger footprint a worthwhile investment. The quantised YOLOv8s model loses less than 1 percentage point of mAP50 compared to the full-precision ONNX model, confirming that quantisation-aware deployment is viable without meaningful accuracy sacrifice.

3.3.3. The critical dimension: actionability

The most important dimension in this comparison is not captured by any single metric: it is the actionability of the model's output. A system's value to a farmer is not measured by its classification accuracy in isolation, but by the quality of the decisions it enables.

Consider the following scenario: a farmer photographs a tractor tyre and the Smart Farmer application processes the image. The MobileNetV2-based system returns: 'Bald Patch Detected. Confidence: 82%.' The YOLOv8-based system returns: 'Bald Patch detected at the outer contact surface (centre of tread), covering approximately 18% of tread area. Confidence: 91%. Sidewall Crack also detected on inner sidewall, minor extent. Confidence: 74%.'

The second output provides the Lifecycle Forecasting Module with four times the information: the presence of two distinct defect types, the spatial location of the primary defect (indicating it is in the highest-stress contact zone, which accelerates the tyre's degradation rate), a quantified defect extent (18% of tread area), and an independent confidence score for each finding. The forecasting model can use the defect area as a direct input to its remaining useful life estimate, triggering a higher-urgency maintenance alert than a tyre on which the bald patch is confined to the less critical shoulder region.

This analysis of actionability conclusively establishes that YOLOv8, despite its greater computational cost, is the appropriate architectural choice for the Computer Vision Wear Detection Module of the Smart Farmer platform.

3.4. Discussion

3.4.1. Significance of the technical evolution

The transition from MobileNetV2 to YOLOv8 documented in this research illustrates a broader principle in the application of computer vision to industrial and agricultural condition monitoring: the right level of model sophistication must be matched to the spatial requirements of the problem. For tasks where the condition of interest manifests as a global, whole-image property for example, distinguishing a photograph of a field with adequate crop cover from one with sparse cover a classifier is the natural and efficient choice. However, for tasks where damage is spatially specific and its location, size, and pattern carry meaningful information, a detector is not simply 'more accurate'; it operates in a fundamentally different and more informative representation space [41].

The literature on automated equipment inspection in industrial settings has drawn similar conclusions. Studies on automated visual inspection of manufacturing components, pipeline welds, and structural elements consistently find that defect localisation via detection or segmentation models provides substantially more useful output for maintenance decision-making than classification-only approaches, even when the classification accuracy is high [42]. This research extends that finding to the agricultural machinery context in a developing-country setting, contributing evidence that the same principle holds when the equipment is simpler and the operating conditions are more variable.

3.4.2. The value of real-world data collection

A second key finding of this research is the demonstrated importance of collecting training data from the actual deployment environment. The decision to gather images from local farms and workshops rather than relying on public datasets or laboratory photography was not merely convenient; it was essential to achieving a model that performs reliably under

real field conditions. The inclusion of images featuring mud, water staining, uneven lighting, cluttered backgrounds, and consumer smartphone camera characteristics in the training data is directly reflected in the model's robustness to these conditions in the test set.

This finding has important methodological implications for future agricultural AI work in developing-country contexts. The 'dataset gap' the absence of locally relevant, contextually appropriate training data is one of the most frequently cited barriers to deploying AI in these settings, yet the solution is not necessarily to wait for large public datasets to emerge. As this research demonstrates, targeted primary data collection from local sources, combined with systematic augmentation to amplify dataset size and diversity, can yield training data of sufficient quality to produce reliable, deployable models. The total dataset size of approximately 1,047 carefully annotated images is modest by the standards of large-scale computer vision research but is entirely adequate for fine-tuning a pre-trained YOLOv8 model to high performance on a defined set of five defect classes.

3.4.3. Limitations of the current system

Despite the strong performance results, the current system has several limitations that should be acknowledged. First, the dataset, while real-world sourced, is limited in geographic diversity. The images were collected from a relatively confined area of Sri Lanka, and the visual characteristics of wear on parts used in different soil types, climates, or farming practices may differ. Second, the model does not currently account for temporal wear progression: it assesses the condition of a part at a single point in time rather than tracking its degradation over a series of inspections. Integration with the Lifecycle Forecasting Module will be needed to realise the full predictive maintenance potential of the system. Third, the model was trained on a limited set of part categories; its ability to assess wear on other components has not been validated.

CHAPTER 4: CONCLUSION AND FUTURE WORK

4.1. Summary of Findings

This report has documented the design, development, and evaluation of the Computer Vision Wear Detection Module the perceptual intelligence layer of the Smart Farmer AI Integrated Spare Part Lifecycle Forecasting System. The research was motivated by the significant economic and safety costs of unmanaged agricultural machinery maintenance in smallholder farming communities, where the lack of automated, accessible, and precise wear detection tools forces farmers into reactive, costly, and sometimes hazardous maintenance practices.

The work progressed through two clearly defined phases, each generating distinct insights that informed the overall research findings.

Phase One demonstrated that MobileNetV2, configured as a transfer-learning image classifier, can achieve competent but limited performance on agricultural spare part wear classification. A test set accuracy of 79.4% was achieved on a seven-class problem covering V-belt and tractor tyre wear conditions. However, detailed analysis of the model's error patterns revealed that this headline accuracy figure concealed fundamental limitations: the inability to localise defects spatially, the inability to detect multiple simultaneous defects in a single image, and poor sensitivity to small localised damage regions. These limitations were not incidental flaws correctable through additional training data or hyperparameter tuning; they were structural properties of the whole-image classification architecture.

Phase Two established that YOLOv8s, configured as a fine-tuned object detection model, resolves all three of the Phase One limitations. The YOLOv8s model achieved a test set mAP50 of 87.1%, a metric that rewards both correct classification and correct spatial localisation simultaneously. The model produces precise bounding boxes around each detected defect exposed threads and surface cracks on V-belts, bald patches, sidewall cracks, and lug wear on tractor tyres at inference times compatible with mobile

deployment. The training accuracy curve demonstrated rapid convergence and strong generalisation, with a training-validation mAP50 gap of only 4.3 percentage points at training completion, indicating that the model had learned general defect-detection representations rather than memorising training-specific patterns.

A foundational contribution of this research, which underpinned the reliability of both model evaluations, was the primary data collection strategy. Images were gathered directly from local farmers and from local iron and engineering workshops rather than from public datasets or laboratory settings. This approach produced a dataset that accurately reflects the visual complexity of real-world agricultural equipment wear in the Sri Lankan context, including the effects of mud, water, uneven lighting, and consumer-grade smartphone photography. The model's demonstrated robustness to these conditions in the test set is a direct consequence of this data collection decision.

In summary, the core findings of this research are:

- Whole-image classification (MobileNetV2) is insufficient for actionable agricultural wear detection due to its lack of spatial awareness, single-label constraint, and poor small-defect sensitivity.
- Object detection (YOLOv8s) is the appropriate architectural choice for this task, providing bounding box localisation, multi-defect detection, and quantifiable defect area output that enables meaningful integration with lifecycle forecasting systems.
- Real-world data collection from local agricultural sources is a necessary and sufficient basis for training a high-performing wear detection model without reliance on unavailable public datasets.
- The YOLOv8s model, after TFLite quantisation, is deployable on mid-range Android devices at an inference latency of approximately 85 ms, which is practical for field use.

4.2. Future Work

4.2.1. Expansion to additional spare part categories

The most natural and impactful extension of this research is the expansion of the wear detection module to cover additional agricultural spare part categories beyond V-belts and tractor tyres. The methodology developed here primary data collection, systematic annotation, YOLOv8 fine-tuning is directly transferable to other part types. Priority candidates for future inclusion include:

- Engine components: Specifically, air filters (characterised by clogging and rupture) and fuel filters (sediment accumulation, discolouration, and membrane damage). Engine-related failures are among the most catastrophic and expensive for smallholder farmers, making early detection particularly valuable.
- Brake pads and brake drums: Wear on brake components has direct safety implications for tractors operating on hilly terrain, and visual inspection of brake pad thickness and drum surface condition is a well-defined detection problem.
- Drive chains and sprockets: Chain wear (elongation and link deformation) and sprocket tooth wear (hooking and undercutting) are visually detectable conditions in agricultural machinery.
- Hydraulic hoses and fittings: Surface cracking, ballooning, and abrasion on hydraulic lines are precursors to rupture and represent a significant risk of sudden machine immobilisation.

Each of these expansions would follow the same workflow: targeted primary data collection from farms and workshops, class-specific annotation guidelines, YOLOv8 fine-tuning, and integration testing with the Lifecycle Forecasting Module. Over time, the accumulated dataset would constitute a comprehensive agricultural spare part wear library specific to the Sri Lankan and regional context.

4.2.2. Instance segmentation for finer wear quantification

The current YOLOv8 implementation uses bounding box detection, which provides a rectangular enclosure around the defect region. While the bounding box area provides a useful first approximation of defect extent, it overestimates the true defect area for irregularly shaped defects like network cracks and lug wear patterns. YOLOv8 also supports instance segmentation, a task in which the model outputs pixel-level masks delineating the exact shape of each detected defect rather than a bounding rectangle [43]. Transitioning to instance segmentation would enable substantially more accurate defect area quantification: the precise number of pixels belonging to a bald patch, for example, could be computed and normalised to the total tyre tread area, yielding a direct percentage-worn metric. This would enhance the lifecycle forecasting module's ability to model non-linear wear progression rates.

4.2.3. Temporal wear tracking

The current module provides a point-in-time wear assessment: it tells the farmer the state of the part as of the moment the photograph was taken. A substantially more powerful capability would be temporal wear tracking: maintaining a longitudinal record of a specific part's wear state across multiple photographic inspections and computing the rate of wear progression between inspections. A part that goes from 5% tread loss to 18% tread loss in two months is wearing at a different rate than a part that shows the same progression over six months, and the forecasting implications are entirely different. Implementing temporal tracking would require a part identification mechanism (to link successive photographs of the same physical part across visits), a database schema for longitudinal wear records, and a regression model to project wear trajectory forward. This would transform the module from a diagnostic tool into a genuinely predictive maintenance system.

4.2.4. Active learning for continuous model improvement

As the Smart Farmer application accumulates users and photographs from a growing diversity of farms and operating conditions, an active learning pipeline would enable the

model to improve continuously from real-world inference. In active learning, the model itself identifies the images on which it is least confident those most likely to be informative for improving its decision boundaries and flags them for human annotation. These newly annotated images are then added to the training set and the model is retrained periodically. This approach has been shown to be highly sample-efficient, allowing models to improve substantially with a relatively small number of new labelled examples [44]. For the wear detection context, an active learning pipeline would be particularly valuable for improving performance on rare defect types (such as severe sidewall cracks) and on novel operating conditions not well represented in the initial training data.

4.2.5. Integration with IoT sensor data

The wear detection module as currently conceived relies on a human to photograph the part and initiate the inspection. A future enhancement would integrate the visual wear assessment with data from low-cost IoT sensors that could trigger automatic inspections. For example, vibration sensors attached to the V-belt drive system can detect the characteristic frequency signature of belt slippage that often accompanies advanced belt wear. A spike in the vibration signal could automatically alert the farmer to photograph the V-belt, or in a fully automated deployment, trigger a camera mounted on the machine to capture an image without any human intervention. Similarly, tyre pressure monitoring systems could flag abnormal pressure loss and trigger a visual inspection request. This sensor-vision fusion approach would reduce reliance on regular human-initiated inspections and move the system closer to fully passive, continuous condition monitoring.

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